# STAFF REVIEW · H446 074 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

## Do Now — source answers

1. Number of items or other measure of problem size.

2. Record of intermediate variables/states.

3. Property maintained through a loop.

4. 128→64→32→16→8→4→2→1; seven halvings.

## I Do — source worked steps

1. Explain — The number of operations doubles when n doubles.

## We Do — guided CHECK

1. Explain
Each iteration cuts remaining size in half.
Original worked example:
O(log n); values 32,16,8,4,2,1: five halvings.

## You Do — source mark schemes

1. O(1) constant [2], O(n) linear [2], O(log n) logarithmic [2].

2. 64→32→16→8→4→2→1 [4], six halvings [2].

3. Binary requires sorted data [2]; preprocessing sort costs time [2]; single lookup may favour linear scan [1], repeated searches justify sorting [1].

## Exit Ticket — source answer

1. False. It describes asymptotic growth.